

# Emily Cook

[emilycook2022@gmail.com](mailto:emilycook2022@gmail.com) | <https://cookemily.github.io/website/index.html>

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## SUMMARY:

Geophysicist experienced in near-surface geophysics, geospatial modeling, energy feasibility, and geohazard assessment. Passionate about community-centric challenges across water, energy, and hazards.

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## EDUCATION:

Binghamton University, SUNY | May 2026 | B.S. in Geophysics (3.93 GPA) | Summa Cum Laude | Dean's List

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## SKILLS:

**Geophysics:** ERT (ResIPy, AGI Earth Imager), SRT (Rayfract), MASW (SurfSeis), GPR (GPRPy), tTEM, HVSr (Grilla), UAV-TIR (Pix4D, Thermoviewer), Borehole geophysics, LiDAR

**Geospatial & Modelling:** ArcGIS Pro, QGIS, Python, Geodatabases, Multi-Criteria Decision Analysis (MCDA)

**Field & Technical:** Survey Design, Technical Writing, UTV/ATV Operation, Backcountry navigation

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## EXPERIENCE:

### Geophysics & Seismology Intern

May 2026 – Present

United States Bureau of Reclamation (BOR)

Denver, CO

- Acquire, process, and invert 2D/3D near-surface geophysical data to characterize canal seepage, spillway voids, and landslide slip planes for federal infrastructure safety.
- Manage and characterize 300+ fault traces, geochronological points, and geophysical surveys in a federal seismic hazard database, streamlining engineering risk analysis.

### Energy Researcher

May 2025 – Present

Alaska Center of Energy and Power (ACEP)

Fairbanks, AK / Remote

- Create and quantify feasible climate solution scenarios of the Energy and Transportation sector for a comprehensive statewide policy report in collaboration with the Nature Conservancy (TNC).
- Collaborate with non-profits, Alaskan stakeholders, and indigenous communities to ensure projects prioritized energy sovereignty, culminating in a published paper and international presentations.
- Led spatial and MCDA for five deepwater Alaskan ports, modeling geological and economic feasibility for geothermal e-fuel production for decarbonization.

### Direct Use Geothermal Researcher

January – December 2024

Summer Scholars & Artists Program

Binghamton, NY

- Built a centralized regional geodatabase, synthesizing depth-to-layer, borehole geophysical logs, and thermal datasets from 1,000+ oil and gas well logs.
- Constructed isopach maps and cross sections to isolate high-probability geothermal potential zones.

### Geology Research Assistant

May – August 2024

SUNY RF funded Basin Research Team

Binghamton, NY

- Data acquisition for unmanned and canoe-borne temperature and hydrogeophysical surveys for submarine groundwater discharge (SGD) in freshwater environments.
- Assisted near-surface geophysical surveys to characterize basin geometry of Green and Round Lakes.

### Drone Geophysics Researcher

August 2022 – January 2024

First-Year Immersion Program (FRI)

Binghamton, NY

- Investigated SGD's role in sustainable water resource management for island communities using UAV-TIR to assess seasonality of seep, summarizing results in research paper and poster session.
- Conducted remote sensing fieldwork and cartography exercises using various geophysical sensors.

## PUBLICATIONS & PRESENTATIONS:

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Cook, E. & de Witt, M. (2025) Assessing the Feasibility of Geothermal-to-X for Sustainable Maritime Refuel Alaska. *Clean Technologies*, 7(4), 115. <https://doi.org/10.3390/cleantechnol7040115>

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|-----------------------------------------------------------------------------------------------------|---------------------------|--------------------|
| Multi-method Geophysical Analysis of West Canal Slope Failure                                       | <i>US BOR</i>             | <b>2026 Poster</b> |
| Geothermal-to-X Potential in Alaska's Maritime Industry                                             | <i>Arctic Frontiers</i>   | <b>2026 Poster</b> |
| <b>*Received Peer-J Best Poster Award and Harpur Edge High Impact Award</b>                         |                           |                    |
| Geothermal-to-X Potential in Alaska's Maritime Industry                                             | <i>ACEP</i>               | <b>2025 Talk</b>   |
| Feasibility of Geothermal-to-X in Alaskan Arctic Shipping                                           | <i>GEODE</i>              | <b>2025 Talk</b>   |
| Spatial Mapping of Geothermal Energy Variables near Binghamton, NY SSAP                             |                           | <b>2024 Talk</b>   |
| Seasonal Variation in Detecting Submarine Groundwater Discharge Using Drone-Mounted Thermal Imaging | <i>FRI Poster Session</i> | <b>2023 Poster</b> |
| Viability of Coastal Seeps to Combat Droughts on Rapa Nui                                           | <i>FRI Poster Session</i> | <b>2022 Poster</b> |

## TECHNICAL PROJECTS:

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1. **Cartographer:** Custom mapping projects (QGIS, ESRI, Leapfrog) and portfolio web pages, including custom LiDAR-art like "Arctic Arteries", which won *Best in Show* at [Art of Science](#) competition (2026).
2. **Geophysics Summer School (Texas A&M):** Seismic refraction field methods; 2018 Kilauea waveform analysis and seismicity using MatLab and QGIS (2025).
3. **GEOL 451 and 453:** Independent course projects in applied geophysical methods. Survey acquisition, processing, and interpretation of wood ash on soil over time (ERT, IP, EMI), locating unmarked graves (GPR), and locating abandoned oil and gas wells (smartphone magnetometers) (2024-2025).
4. **Rover Team Geology Lead:** Integration of remote sensing and hydrologic interpretation for autonomous planetary analog missions (2023-2024).
5. **Ecology Field Courses:** Immersive geological and ecological field study in the Everglades and Utah and Wyoming National Parks over five weeks, awarded 'Outside the Classroom' (EF) Award (2023).

## LEADERSHIP & OUTREACH:

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1. **Guest Science Educator:** Designed and delivered interactive climate and energy presentations connecting the Arctic issues with local issues for K-12 students (2025-2026).
2. **Team Captain (Binghamton Ultimate Frisbee):** Led and managed logistics for ~35 athlete roster for travel and competitive play across East Coast; two National Championship appearances (2024-2026).
3. **Ralph Digman Award** – Outstanding Binghamton University Earth Science Senior (2026).
4. **Phi Beta Kappa** – Binghamton University Chapter (2026).
5. **SUNY Impact Scholarship** — Awarded by the New York State Wireless Association (2024).